



Isaac Biology Resources

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Session outline



- Overview of Isaac Biology resources
- A Level book
- GCSE questions

Session outline



- **Overview of Isaac Biology resources**
- A Level book
- GCSE questions

Overview of Isaac Biology resources



 You work it out.

About IsaacQuestion finderConceptsNewsEvents ▾BooksHelp ▾⚙ ▾Log out

LTMy Isaac

Explore by learning stage

11-14GCSEA LevelUniversity

Explore by subject

PhysicsMathsChemistryBiology

Search

Welcome back, L. Thomson!

GCSE Biology

A Level Biology

Dashboard view

TeacherStudent

View group progress

Biology Tutorial Phase 2 2026

B4 Biology Phase 2 2026

B3 Biology Phase 2 2026

B2 Biology Phase 2 2026

B1 Biology Phase 2 2026

[See all groups' progress](#)

View scheduled work

 **STEM SMART Biology Week 20 - Circulatory Systems 2**
Due in 4 daysSTEM SMART 2026 Phas...

 **STEM SMART Biology Week 21 - Plant Growth & Reproduction**
Due on Sun, 12 Jul 2026STEM SMART ...

 **STEM SMART Biology Week 22 - Plant Nutrient & Water**

[See assignment schedule](#)[See all set tests](#)

Explore our books

All ▾


Step into Physics


Step Up to GCSE Physics


Essential GCSE Physics

[See all books](#)

More in My Isaac

Teacher features

Manage groups

Create a question deck

Set assignments

Assignment schedule

Assignment progress

Set / manage tests

My account

Biology homepages



 **GCSE Biology**

Try a random question! [Get a different question](#)

 **Membrane Transport Examples** 

Biology | Cell Biology | Cell Transport

GCSE Practice 1 

 Question finder Find GCSE Biology questions to try by topic and difficulty level. Find questions	 Question decks by topic Practise specific topics by using our ready-made question decks. View topic question decks
 Glossary Use the glossary to understand the vocabulary you need for GCSE Biology. Browse the glossary	 Core skills practice Practise core maths skills that are used in GCSE Biology. Practise mental maths

 **A Level Biology**

Try a random question! [Get a different question](#)

 **Haemoglobin and Oxygen Transport** 

Biology | Physiology | Exchange & Transport

A Level Practice 2 

 Question finder Find A Level Biology questions to try by topic and difficulty level. Find questions	 Question decks by topic Practise specific topics by using our ready-made question decks. View topic question decks
 Concepts Review the key concepts for A Level Biology. Explore concepts	 Glossary Use the glossary to understand the vocabulary you need for A Level Biology. Browse the glossary
 Tests Use tests to practise a range of topics. These tests are available for you to freely attempt. Find a test	 Biology extension Stretch your understanding of biology with our extension questions that make you think outside the box. View extension questions

Question finder



A Level Biology >

Question finder

The questions shown on this page have been filtered to only show those that are relevant to A Level Biology. You can browse all questions [here](#).

Search questions

Filter questions by

Learning Stage

☐ A Level

☐ Further A

Topic

☐ Cell Biology

☐ Biochemistry

☐ Genetics

☐ Physiology

☐ Ecology

☐ Evolution

☐ Maths Skills

Showing 30 of 362.

Shuffle questions

A Bacterial Planet
Biology | Cell Biology | Cell Division

A Level Challenge 3
Further A Challenge 1

A Biological Rubik's Cube: Levinthal's Paradox of Protein Folding
Biology | Biochemistry | Biomolecules

A Level Challenge 3
Further A Practice 3

ABO Blood Types & Hardy-Weinberg Equilibrium
Biology | Evolution | Theory & Evidence

A Level Challenge 3

ATP and NAD
Biology | Biochemistry | Bioenergetics

A Level Practice 3

Action Potentials
Biology | Physiology | Sense & Movement

A Level Practice 3

Active Transport
Biology | Cell Biology | Cell Transport

A Level Practice 1

Questions: fields & topics



☒ Cell Biology

- ☐ Cell Structure
- ☐ Cell Division
- ☐ Cell Transport
- ☐ Cell Specialisation

☒ Biochemistry

- ☐ Biomolecules
- ☐ Enzymes
- ☐ Bioenergetics

☒ Genetics

- ☐ Nucleic Acids
- ☐ Protein Synthesis
- ☐ Inheritance
- ☐ Biotechnology

☒ Physiology

- ☐ Nutrition & Digestion
- ☐ Exchange & Transport
- ☐ Homeostasis & Response
- ☐ Sense & Movement
- ☐ Health & Disease
- ☐ Reproduction & Development

☒ Ecology

- ☐ Populations
- ☐ Ecosystems
- ☐ Biodiversity

☒ Evolution

- ☐ Variation
- ☐ Theory & Evidence
- ☐ Speciation
- ☐ Classification

☒ Maths Skills

- ☐ Statistical Tests

Questions: difficulty



Difficulty



[Learn more about difficulty levels](#)

☐ Practice 1

☐ Practice 2

☐ Practice 3

☐ Challenge 1

☐ Challenge 2

☐ Challenge 3

- Practice:
 - Recall of information, basic calculations
 - Common question types
 - Drag & drop
 - Short answer (1 or 2 word)
 - Reorder
- Challenge
 - Application, logic & reasoning, problem-solving
 - Common question types
 - Item (multiple choice with multiple correct options)
 - Calculation

Question: learning stage



Learning Stage

☐ A Level

☐ Further A

- More than just A Level biology required
- May require A Level maths or chemistry



This question involves using exponentials and logarithms , which are part of A Level Maths. For more information please check with your teacher.

Question decks by topic



Question decks by topic

Decks by stage

GCSE

A Level

Decks by subject

Physics

Chemistry

Maths

Biology

Teacher notes >

A Level Biology

The following decks use questions from across the website, and are ideal to use in lessons, for homework or when revising. The decks are not comprehensive, i.e. they do not contain all the Isaac Biology questions on the topic. More questions can often be found using the [Question Finder](#).

Jump to section

Biochemistry

Cell Biology

Ecology

Evolution

Genetics

Physiology

Statistics

Biochemistry >

Cell Biology >

Ecology >

Evolution >

Genetics >

Concept pages (A Level only)



Search concepts



Filter by topic

All 8

Cell Biology 1

Biochemistry 1

Evolution 2

Maths Skills 4

The concepts shown on this page have been filtered to only show those that are relevant to A Level Biology. You can browse all concepts [here](#).



Showing 8 results



An Introduction to Statistics in Biology

Why we use statistical tests, when to use each test, and how to interpret the results.



Carbohydrates

The structures and functions of monosaccharides, disaccharides, polysaccharides



Chi-squared Tests in Biology

When and how to use chi-squared tests in biology.



Phylogenetic Trees

How to interpret phylogenetic trees



Spearman's Rank Correlation Coefficient in Biology

When and how to use Spearman's rank correlation coefficient in biology.



Student's t-test (Unpaired Samples) in Biology

When and how to use Student's t-test (unpaired samples) in biology.



The Eukaryotic Cytoskeleton

The structures that make up the eukaryotic cytoskeleton and their functions in the cell



The Hardy-Weinberg Principle

What the Hardy-Weinberg principle is, and how to use it to calculate expected genotype frequencies.

Glossary



A Level Biology >



A Level Biology Glossary

Use our glossary to find definitions of important words and phrases.



Search glossary

e.g. Cell



Switch learning stage

GCSE

A Level

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

#

5' cap

A methylated guanine nucleotide connected to the 5' end of an mRNA molecule via a 5' to 5' triphosphate linkage.

A

Active transport

The movement of ions/molecules across a membrane up/against their concentration gradient (i.e. from a low concentration to a high concentration).

ADP

(adenosine diphosphate) A nucleotide composed of the nitrogenous base adenine, a pentose sugar, and two phosphate groups. Formed by the hydrolysis of ATP in a reaction that also produces inorganic phosphate and releases energy.

Tests (A Level only)



A Level Biology >

 **Practice tests**

This page lists tests you can take whenever you'd like to practise your skills and check your understanding.

Search practice tests



You can see all of the tests that you have in progress or have completed in your My Isaac:

[My tests →](#)

	Biology Admissions Practice 1 Visible to students	Preview	Set test →
	Biology Admissions Practice 2 Visible to students	Preview	Set test →
	Biology Admissions Practice 3 Visible to students	Preview	Set test →
	Biology Admissions Practice 4 Visible to students	Preview	Set test →
	Biology Admissions Practice 5 Visible to students	Preview	Set test →
	Biology Admissions Practice 6 Visible to students	Preview	Set test →
	Biology Admissions Practice 7 Visible to students	Preview	Set test →

Extension questions (A Level only)



 **Biology Extension Questions**

[Go to homepage](#) →

On this page you can find A Level (or equivalent) questions that will help you think about biological topics on a deeper level, by exploring interesting questions like 'Why don't humans photosynthesise?' and 'How fast would ET run?'. Each question is split into multiple parts, with each part building on the previous part to help you gradually think through and answer the overall question.

These questions were written by Dr Andrew Catherall-Ostler, supported by the Isaac Newton Trust.

The questions are grouped by field. You can use the links below to jump to the questions within that field.

Jump to section

[Biochemistry](#) [Cell Biology](#) [Ecology](#) [Genetics](#)

[Physiology](#) [Mathematical Biology](#)

[Biochemistry >](#)

[Cell Biology >](#)

[Ecology >](#)

[Genetics >](#)

[Physiology >](#)

Session outline



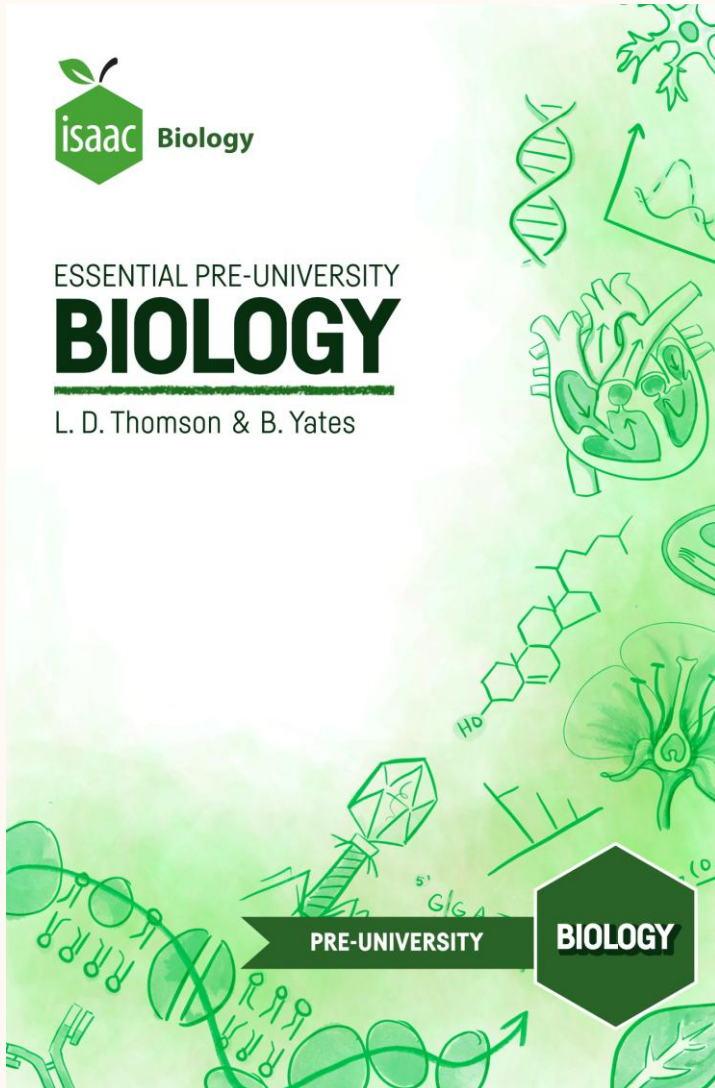
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- A Level book
- GCSE questions

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A Level biology book



- Compilation of (updated) A Level biology questions from the site
- Release early 2027
- 9 chapters
 - A-H: Biology chapters
 - I: Statistics
- Can be used standalone or in conjunction with the online book (free).



tinyurl.com/isaacbiologybook

A Level biology book



B1.3 Identifying Organelles 1

A Level P1 ●●●

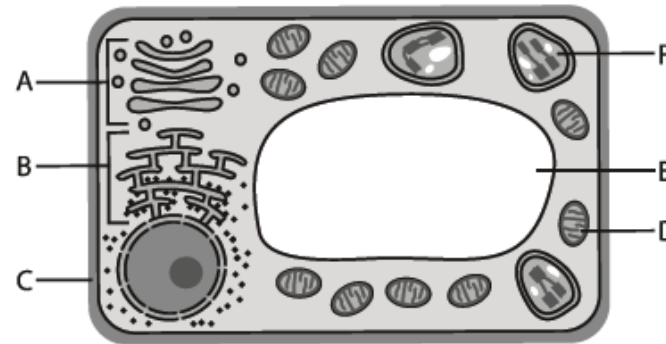


Figure 31 A diagram of a cell.

a) What kind of cell is shown above? Write down the correct answer from the list below.

- ☐ Animal cell
- ☐ Plant cell
- ☐ Bacterial cell

b) Match the labels in Figure 31 to the organelles in the list below.

- cell wall
- centriole
- chloroplast
- endoplasmic reticulum
- flagellum
- Golgi apparatus
- mitochondrion
- vacuole

c) What is organelle C made of?

d) What is the name of the membrane that surrounds organelle E? Write down the correct answer from the list below.

- ☐ capsule
- ☐ cisternae
- ☐ tonoplast
- ☐ cell membrane
- ☐ inner membrane

e) What is the primary role of organelle F?

A Level biology book



A1.1 Monosaccharides

A Level P1 ●●●

Monosaccharides are the smallest carbohydrates, and are the monomers from which larger carbohydrates are made. Monosaccharides are categorised based on how many carbon atoms they contain, as follows:

- **triose:** three carbons
- **tetrose:** four carbons
- **pentose:** five carbons
- **hexose:** six carbons
- **heptose:** seven carbons

- a) Name two pentose monosaccharides.
- b) Name three hexose monosaccharides.
- c) Write down the chemical formula for a hexose monosaccharide.

A Level biology book



A3.2 Identifying Dipeptides

A Level P1 ●●●

a) Draw two dipeptides that could be produced by a condensation reaction between the amino acids cystine and valine, shown below in Figure 13.

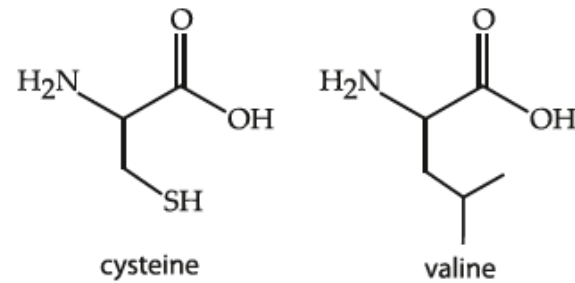


Figure 13

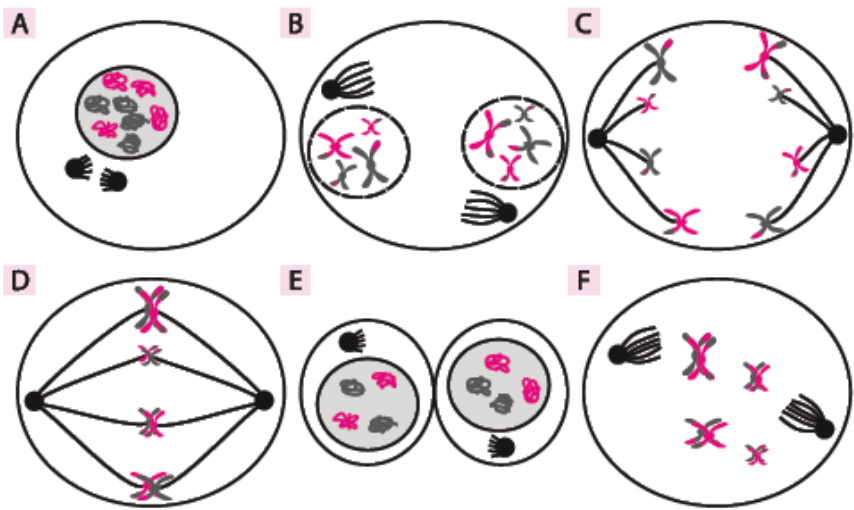
A Level biology book



B2.6 Stages of Meiosis

A Level P1 ●●●

a) The images below represent different stages of meiosis I, for an organism with a diploid chromosome number of 8.



Redraw and fill in the table below to put the stages above in the correct order, and write the name for each stage.

Image	Stage
A	interphase
E	cytokinesis

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Session outline

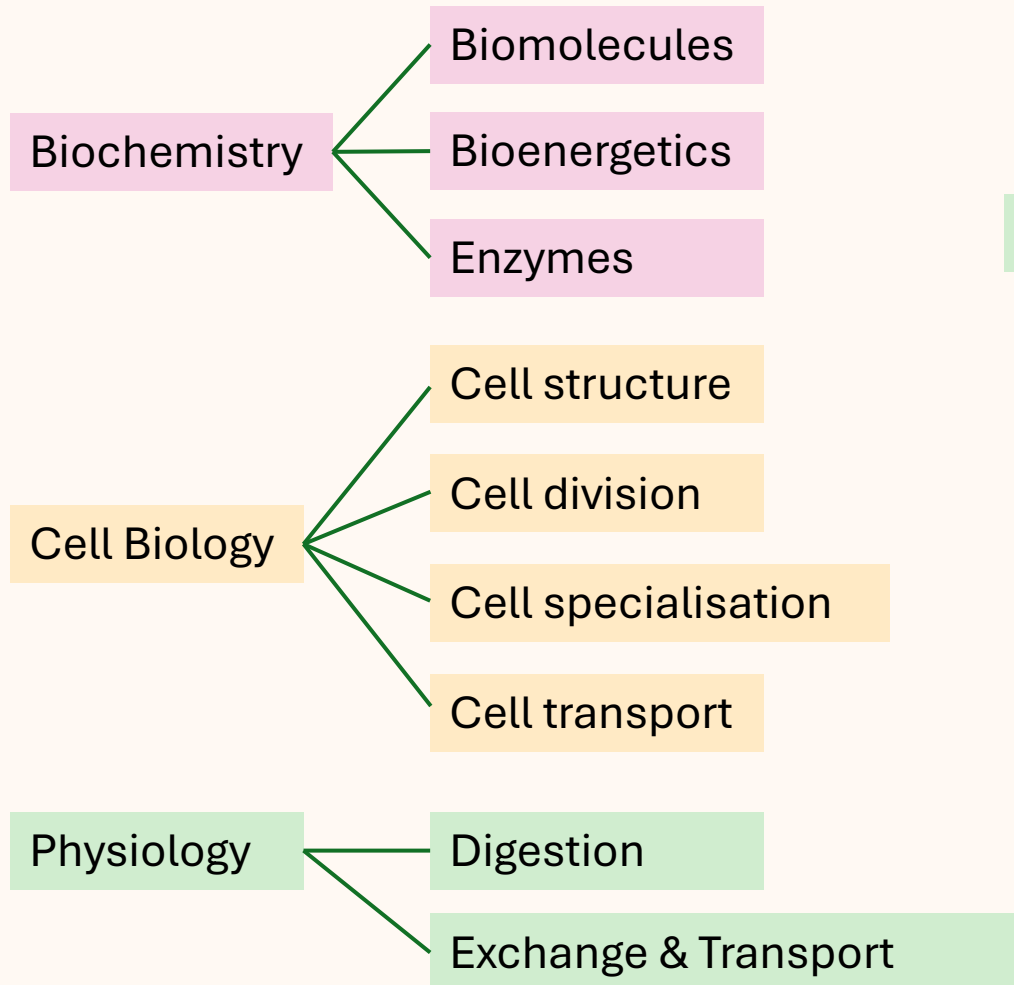


- Overview of Isaac Biology resources
- A Level book
- **GCSE questions**

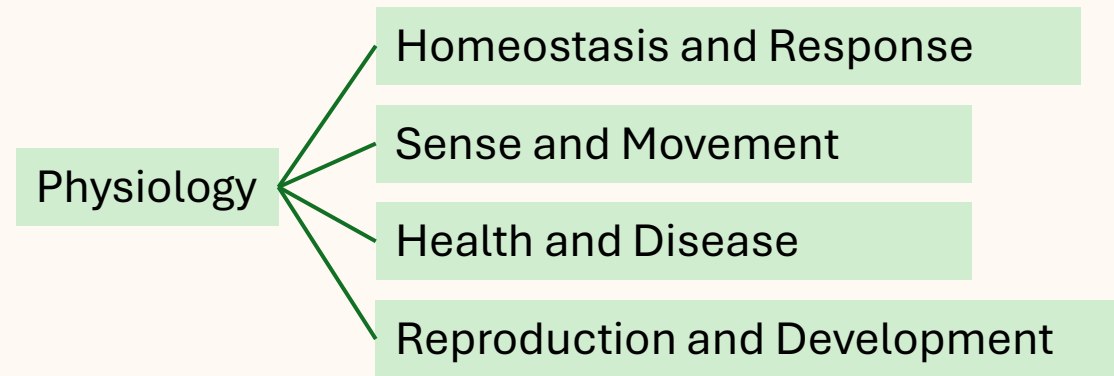
GCSE Questions



Available now



Available by end of 2026



Available by start of Autumn term 2027

Ecology

Genetics

Evolution

Types of GCSE Practice Questions



Recall

Part A
Animal cell ▾

Label the organelles and structures in this diagram of an animal cell.

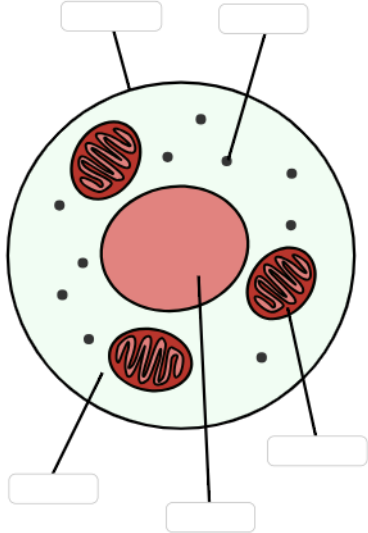


Figure 1: An animal cell.

Check my answer

Diagnostic

Part A
Nature ▾

Which of the following statements about enzymes are **true**? Select all that apply.

- ☐ Enzymes are microorganisms
- ☐ Enzymes are alive.
- ☐ Very high temperatures kill enzymes.
- ☐ Enzymes are proteins.
- ☐ Enzymes are cells.

Check my answer

Investigative

A class of students investigated osmosis using pieces of carrot. They prepared four solutions, each containing a different concentration of sucrose (a type of sugar). They measured the initial mass of their pieces of carrot, then left one piece of carrot in each solution, or distilled water, for 24 hours. They removed the carrot pieces from the solutions, dried them and measured their mass again. They then calculated the change in mass (in g) and the percentage (%) change in mass.

Part A
Completing the data table ▾

The table below contains some of the measurements from the experiment. Fill in the missing measurements. Round all values to 1 decimal place.

Sucrose concentration (mol/dm ³)	Initial mass (g)	Final mass (g)	Change in mass (g)	Change in mass (%)
0.0	5.0	5.4	0.4	8.0
0.2	5.1	5.2		
0.4	6.2			-1.6
0.6		5.7	-0.2	
0.8	5.0	4.8		

Check my answer

Practice vs Challenge Comparison – Meiosis



P1 Meiosis Fundamentals

The question reinforces **subject knowledge**:

- The stages of meiosis

This question can be used:

- as a diagnostic to check students' basic understanding
- to help students consolidate their learning through recall

C3 Meiosis Variations

The question reinforces **subject knowledge**:

- The stages of meiosis
- Genetics and inheritance: sex chromosomes, genetic disorders, genetic variation

The question develops **skills**:

- Reading comprehension
- Logical reasoning

This question can be a jumping off point for class discussions.

Try out some GCSE biology questions!



https://isaacscience.org/question_decks#ists26_sat_2c_gcse_bio



ISTS 26: GCSE Biology

Cell Labelling	In progress	GCSE Practice 1	
Mystery Cell		GCSE Challenge 1	
Enzyme Statements		GCSE Practice 1	
Enzyme Deficiency		GCSE Challenge 1	
Meiosis Fundamentals		GCSE Practice 1	
Meiosis Variations		GCSE Challenge 3 A Level Challenge 1	
Osmosis Investigation		GCSE Practice 1	
Differential Diffusion		GCSE Challenge 3	
Goldfish Metabolism		GCSE Challenge 2	
Algae Balls		GCSE Challenge 2	

Try out some GCSE biology questions!



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